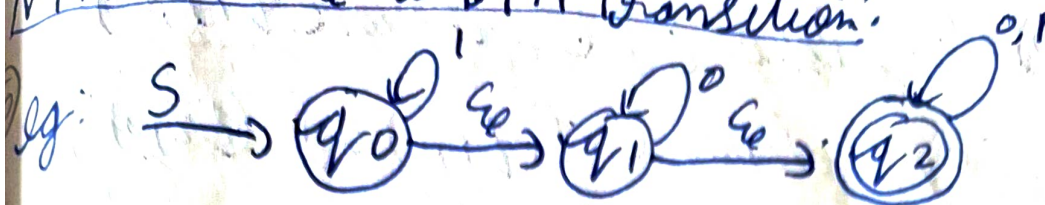


NFA with ϵ to DFA Transition:



NFA Transition Table:

state \ IP	0	1	ϵ
q_0	—	$\{q_0\}$	$\{q_1\}$
q_1	$\{q_1\}$	—	$\{q_2\}$
q_2	$\{q_2\}$	$\{q_2\}$	—

$$\epsilon\text{-closure}(q_0) = \{q_0, q_1, q_2\}$$

$$\epsilon\text{-closure}(q_1) = \{q_1, q_2\}$$

$$\epsilon\text{-closure}(q_2) = \{q_2\}$$

$$\begin{aligned} \delta'(q_0, 0) &= \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_0), 0)) \\ &= \epsilon\text{-closure}(\delta(q_0, 0) \cup \delta(q_1, 0) \cup \delta(q_2, 0)) \\ &= \epsilon\text{-closure}(\{q_0, q_1, q_2\}) \end{aligned}$$

$$\delta'(q_0, 0) = \{q_0, q_1, q_2\}$$

$$\begin{aligned} \delta'(q_0, 1) &= \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_0), 1)) \\ &= \epsilon\text{-closure}(\delta(q_0, 1) \cup \delta(q_1, 1) \cup \delta(q_2, 1)) \\ &= \epsilon\text{-closure}(\{q_0, q_2\}) \end{aligned}$$

$$\delta'(q_0, 1) = \{q_0, q_2\}$$

$$\begin{aligned} f'(q_1, 0) &= \epsilon\text{-closure}(f(\epsilon\text{-closure}(q_1), 0)) \\ &= \epsilon\text{-closure}(f(q_1, 0) \cup f(q_2, 0)) \\ &= \epsilon\text{-closure}(q_2) \end{aligned}$$

$$f'(q_1, 0) \Rightarrow \{q_2\} \text{ —}$$

$$\begin{aligned} f'(q_1, 1) &= \epsilon\text{-closure}(f(\epsilon\text{-closure}(q_1), 1)) \\ &= \epsilon\text{-closure}(f(q_1, 1) \cup f(q_2, 1)) \\ &= \epsilon\text{-closure}(q_2) \end{aligned}$$

$$f'(q_1, 1) \Rightarrow \{q_2\} \text{ —}$$

$$\begin{aligned} f'(q_2, 0) &= \epsilon\text{-closure}(f(\epsilon\text{-closure}(q_2), 0)) \\ &= \epsilon\text{-closure}(f(q_2, 0)) \\ &= \epsilon\text{-closure}(q_2) \end{aligned}$$

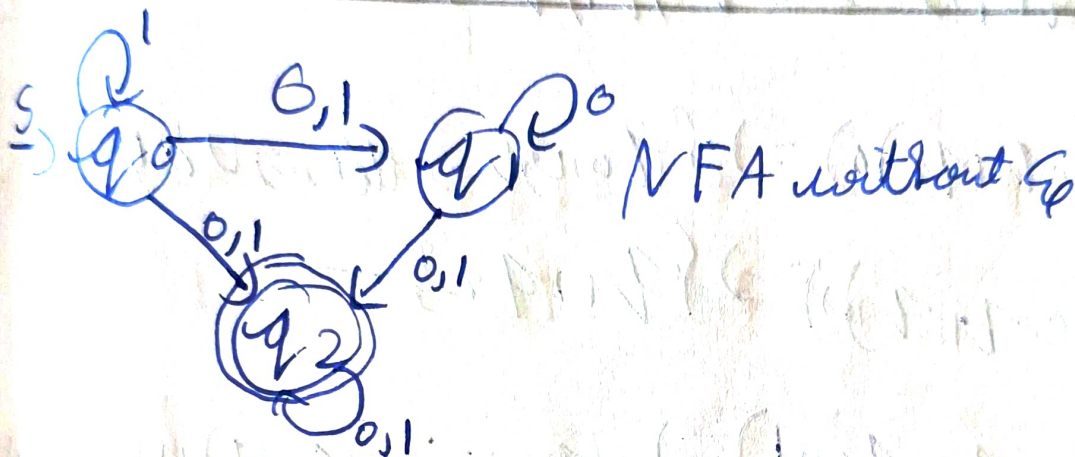
$$f'(q_2, 0) \Rightarrow \{q_2\} \text{ —}$$

$$\begin{aligned} f'(q_2, 1) &= \epsilon\text{-closure}(f(\epsilon\text{-closure}(q_2), 1)) \\ &= \epsilon\text{-closure}(f(q_2, 1)) \\ &= \epsilon\text{-closure}(q_2) \end{aligned}$$

$$f'(q_2, 1) \Rightarrow \{q_2\} \text{ —}$$

Transition Table: NFA without ϵ :

State \ IP	0	1
q_0	$\{q_1, q_2\}$	$\{q_0, q_1, q_2\}$
q_1	$\{q_1, q_2\}$	$\{q_2\}$
q_2	$\{q_2\}$	$\{q_2\}$



NFA without ϵ to DFA:

DFA Transition Table

State \ IP	0	1
$\{q_2\}$	$\{q_1, q_2\}$	$\{q_0, q_1, q_2\}$
$\{q_1, q_2\}$	$\{q_1, q_2\}$	$\{q_0, q_1, q_2\}$
$\{q_0, q_1, q_2\}$	$\{q_1, q_2\}$	$\{q_0, q_1, q_2\}$

$$\delta(q_0, q_1, 0) \Rightarrow \delta(q_0, 0) \cup \delta(q_1)$$

$$\delta(q_0, q_1, 0) \Rightarrow \{q_1, q_2\} -$$

$$\delta(q_0, q_1, 1) \Rightarrow \delta(q_0, 1) \cup \delta(q_1, 1)$$

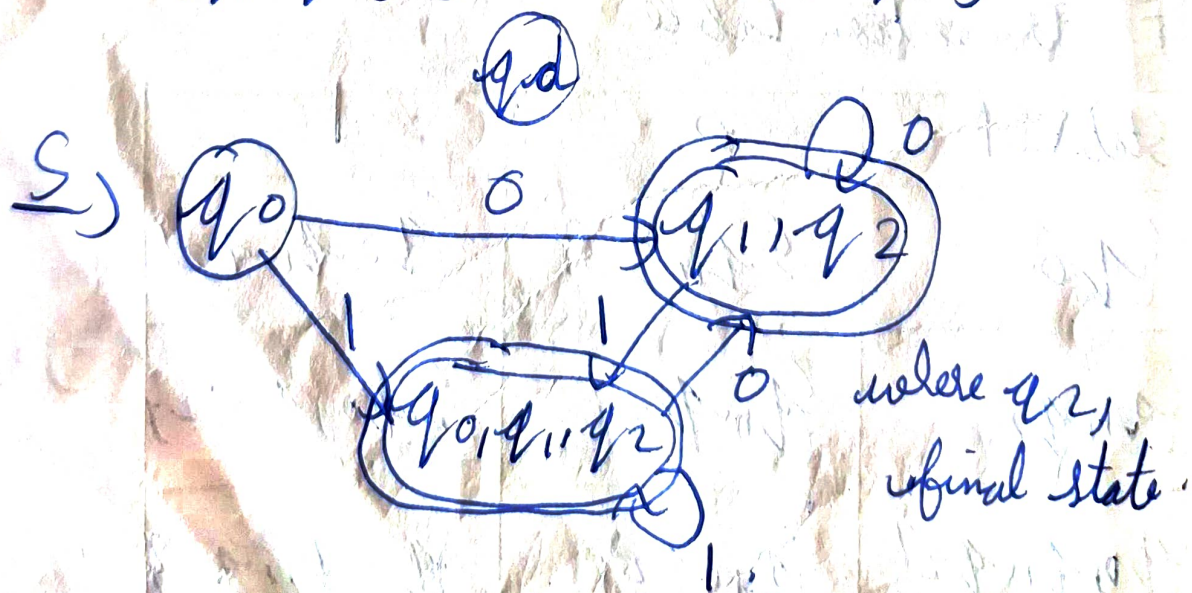
$$\delta(q_0, q_1, 1) \Rightarrow \{q_0, q_1, q_2\} -$$

$$\delta(q_0, q_1, q_2, 0) \Rightarrow \delta(q_0, 0) \cup \delta(q_1, 0) \cup \delta(q_2, 0)$$

$$\delta(q_0, q_1, q_2, 0) \Rightarrow \{q_1, q_2\} -$$

$$\delta(q_0, q_1, q_2, 1) \Rightarrow \delta(q_0, 1) \cup \delta(q_1, 1) \cup \delta(q_2, 1)$$

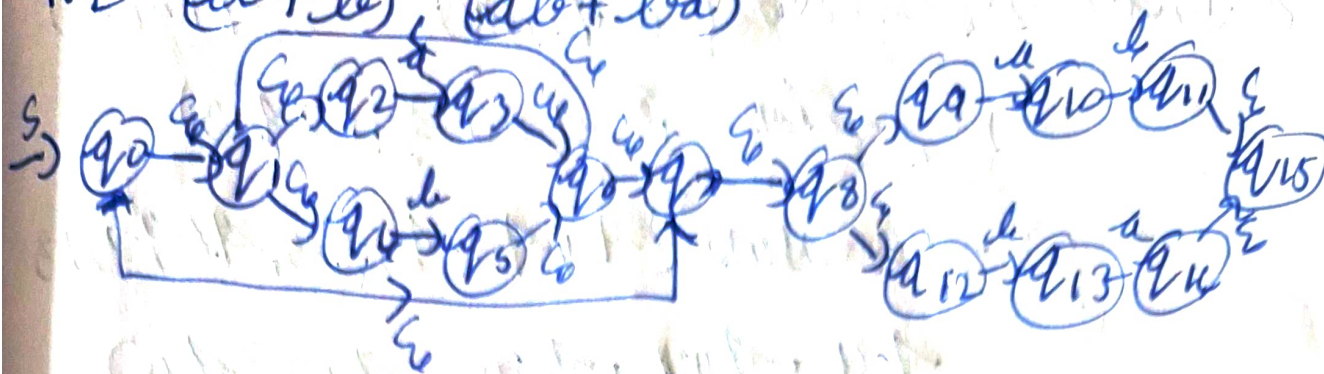
$$\delta(q_0, q_1, q_2, 1) \Rightarrow \{q_0, q_1, q_2\} -$$



q_d is dead state is ϕ present in transition table.

2) NFA with ϵ to DFA directly:

$$RE = (a + b)^* (ab + ba)$$



$$\epsilon\text{-closure}(q_0) \Rightarrow \{q_0, q_1, q_2, q_4, q_7, q_8, q_9, q_{12}\} \Rightarrow A$$

$$\epsilon\text{-closure}(A) \Rightarrow$$

$$f(A, a) \Rightarrow \epsilon\text{-closure}(f(\epsilon\text{-closure}(A), a))$$

$$\Rightarrow \epsilon\text{-closure}(f(q_0, a) \cup f(q_1, a), \dots \cup f(q_{12}, a))$$

$$\Rightarrow \epsilon\text{-closure}(\{q_3, q_{10}\})$$

$$\Rightarrow \{q_1, q_2, q_3, q_4, q_6, q_7, q_8, q_9, q_{10}, q_{12}\} \Rightarrow B$$

$$f(A, b) \Rightarrow \epsilon\text{-closure}(f(\epsilon\text{-closure}(A), b))$$

$$\Rightarrow \epsilon\text{-closure}(\{q_5, q_{13}\})$$

$$\Rightarrow \{q_1, q_2, q_4, q_5, q_6, q_7, q_8, q_9, q_{12}, q_{13}\} \Rightarrow C$$

$$f(B, a) \Rightarrow \epsilon\text{-closure}(f(q_1, a), f(q_2, a), \dots, f(q_{12}, a))$$

$$\Rightarrow \epsilon\text{-closure}(\{q_3, q_{10}\}) \Rightarrow B$$

$$f(B, b) \Rightarrow \epsilon\text{-closure}(f(q_1, b) \dots f(q_{12}, b))$$

$$\Rightarrow \epsilon\text{-closure}(q_5 \cup q_{11} \cup q_{13})$$

$$\Rightarrow \epsilon\text{-closure}(q_1, q_2, q_4, q_5, q_6, q_7, q_8, q_9, q_{11}, q_{12}, q_{13}, q_{15})$$

$$\Rightarrow D$$

$$f(C, a) \Rightarrow \epsilon\text{-closure}(f(\epsilon\text{-closure}(C), a))$$

$$\Rightarrow \epsilon\text{-closure}(f(q_1, a) \cup f(q_{12}, a))$$

$$\Rightarrow \epsilon\text{-closure}(q_3 \cup q_{10} \cup q_{14})$$

$$\Rightarrow \{q_1, q_2, q_3, q_4, q_6, q_7, q_8$$

$$q_9, q_{10}, q_{12}, q_{14}, q_{15}\}$$

$$\Rightarrow E$$

$$f(C, b) \Rightarrow \epsilon\text{-closure}(f(\epsilon\text{-closure}(C), b))$$

$$\Rightarrow \epsilon\text{-closure}(f(q_1, b) \cup f(q_{12}, b))$$

$$\Rightarrow \epsilon\text{-closure}(q_5 \cup q_{13})$$

$$\Rightarrow C$$

$$\begin{aligned}
 f(Q, a) &\Rightarrow \epsilon\text{-closure}(f(\epsilon\text{-closure}(Q), a)) \\
 &\Rightarrow \epsilon\text{-closure}(f(q_1, a) \cup \dots \cup f(q_{15}, a)) \\
 &\Rightarrow \epsilon\text{-closure}(q_3 \cup q_{10} \cup q_{14}) \\
 &\Rightarrow E -
 \end{aligned}$$

$$\begin{aligned}
 f(Q, b) &\Rightarrow \epsilon\text{-closure}(f(q_1, b) \cup \dots \cup f(q_{15}, b)) \\
 &\Rightarrow \epsilon\text{-closure}(q_5 \cup q_{13}) \\
 &\Rightarrow C -
 \end{aligned}$$

$$\begin{aligned}
 f(E, a) &\Rightarrow \epsilon\text{-closure}(f(q_1, a) \cup \dots \cup f(q_{15}, a)) \\
 &\Rightarrow \epsilon\text{-closure}(q_3 \cup q_{10}) \\
 &\Rightarrow A -
 \end{aligned}$$

$$\begin{aligned}
 f(E, b) &\Rightarrow \epsilon\text{-closure}(f(q_5, b) \cup \dots \cup f(q_{13}, b)) \\
 &\Rightarrow D -
 \end{aligned}$$

Transition Table DFA:

State \ IP	a	b
A A	A B	C
B B	A B	D
C C	E	C
D D	E	C
E E	A	D

